Pipes & Cisterns Booklet Questions

Ex:-23:- [B.Q:-1]:- A & B can fill the container in 8 & 12 hr's. & another pipe 'C' can empty it in **6** hr's. If all the pipes are opened together, then the container will be filled in?

Ans:- $A = \frac{1}{8}$, $B = \frac{1}{12}$, $C = \frac{-1}{6}$

Comb eff $= \frac{1}{8} + \frac{1}{12} - \frac{1}{6} = \frac{3+2-4}{24} = \frac{1}{24}$

Total No. of hr's $= \frac{1}{1/24} = 24$ hr's to fill the tank

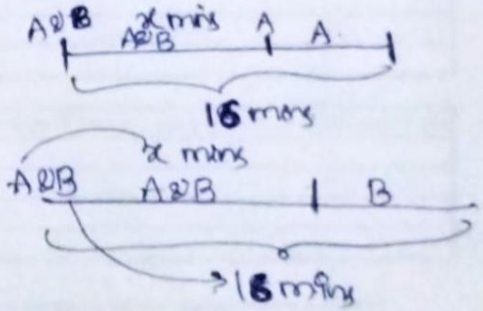
[Because result is +ve sign]

(or).

Units method:- 8, 12, −6

$$\frac{24}{\frac{24}{8} + \frac{24}{12} - \frac{24}{6}} = \frac{24}{3+2-4} = 24 \text{ hr's to fill.}$$

Ex:-23:- [B.Q:-2]:- Two pipes A & B can fill a cistern in 18 mins & 27 mins. If both the pipes are opened & A is turned off after 'x' minutes. then the cistern will be filled in 16 mins. Find x?



Ans:- $A = \frac{1}{18}$, $B = \frac{1}{27}$

$mA + nB = 1$

$x \times \frac{1}{18} + 16 \times \frac{1}{27} = 1$

$\frac{3x + 2 \times 16}{54} = 1 \Rightarrow 3x = 54 - 32 = 22$

$\Rightarrow [x = 22/3 = 7\frac{1}{3} \cong 7 \text{ mins}]$

Ex:-24:- [B.Q:3]:- 3 pipes P & Q can fill in 5 & 6 hr's & R can empty in **2** hr's a cistern. If Pipes are opened in order at 7am, 8am & 9am., then at what time will the cistern be emptied?

(15)

Ans:- Pipe A @ 7am [7am-8am] = 1hr = $\frac{1}{5}$ of work

1st hr Pipe A ⇒ $\frac{1}{5}$ of tank filled.

2nd hr ⇒ Pipe B @ 8am [8am-9am] = 1hr work by both A & B because A already started @ 7am

2nd hour work = $\frac{1}{5}+\frac{1}{6} = \frac{6+5}{30} = \frac{11}{30}$ tank filled

⇒ In 1st 2hrs tank filled = $\frac{1}{5}+\frac{11}{30} = \frac{6+11}{30} = \frac{17}{30}$ of tank.

3rd hour ⇒ Pipe C @ 9am joined [9am-10am] = 1hr work. both A, B & C because A & B already started.

3rd hour work = $\frac{1}{5}+\frac{1}{6}-\frac{1}{2} = \frac{6+5-15}{30} = \frac{-4}{30}$ of tank emptied.

⇒ means after 9am every hr 3 pipes are working & emptying the filled tank from 9am.

Filled tank $\left[\frac{17}{30}\right]$ to be emptied from 9am for every hr $\frac{4}{30}$ of tank.

$\frac{4}{30}$ ⟶ emptied in ⟶ 1hr

$\frac{17}{30}$ ⟶ ?

$= \frac{\frac{17}{30}\times 1}{\frac{4}{30}} = \frac{17}{4} = 4\frac{1}{4}$ hrs = 4 hrs 15 mins

from 9am emptying start & that

9am + $4\frac{1}{4}$ hrs = 9am + 4:15 mins

= 1:15 pm tank emptied.

Ex:-25:- [B.Q:-5]:- A, B, C together can fill an empty tank in 6 days. Pipe B is twice as fast as A & C is thrice as fast as B. Find the time taken by A & B to fill the tank?

$A+B+C = \frac{1}{6}$ & $B = 2A$, $C = 3B$

Converting all in terms of A $\Rightarrow B = 2A$, $C = 3\times 2A = 6A$

$A+2A+6A = \frac{1}{6} \Rightarrow 9A = \frac{1}{6} \Rightarrow A = \frac{1}{54} \Rightarrow A \to 54$ days

$B = 2A = 2\times\frac{1}{54} = \frac{1}{27} \Rightarrow B \to 27$ days.

together $= \frac{54\times 27}{54+27} = \frac{18}{\cancel{81}\times 27} = 18$ days.

Ex:-26:- [B.Q:-6]:- $A = \frac{1}{12}$, $B = \frac{1}{16}$. Both opened together @ 10am. & B is turned off after some time, then tank filled @ 7 pm. At what time B turned off?

Ans:- work done hr's = 7Pm − 10am = 9 hr's.

'A' worked for total 9 hr's.

Let's assume 'B' turned off after x hr's.

$9\times\frac{1}{12} + x\times\frac{1}{16} = 1 \Rightarrow \frac{4\times 9 + 3\times x}{48} = 1.$

$3x = 48 - 36 = 12 \Rightarrow \boxed{x = 12/3 = 4 \text{ hr's}}$

after 4 hr's to 10am B turned off.

So, turning off time is 10am + 4hr's = (2 pm)

Ex:-27:- [B.Q:-7]:- A pipe can fill a tank in 4 hr's. A leakage @ bottom of the tank emptying 2 ltrs per minute. Because of leak it's taking $\frac{1}{2}$ hr more than the usual time. What is capacity of the tank?

(7)

Ans:- $A = \frac{1}{4}$, due to B(leak) $\frac{1}{2}$ hr' extra

means $A+B = \frac{1}{4\frac{1}{2}} = \frac{1}{9/2} = \frac{2}{9}$

Assume B in x hrs to empty means $B = \frac{-1}{x}$

$\frac{1}{4} - \frac{1}{x} = \frac{2}{9} \Rightarrow \frac{x-4}{4x} = \frac{2}{9}$

$9(x-4) = 8x \Rightarrow 9x - 8x = 36 \Rightarrow x = 36$ hr's to empty

36 hr's in minutes $= 36 \times 60 = 2160$ mins

Capacity of the tank $= 2160 \times 2$ ltrs (2 ltrs in 1 minute)

$= 4320$ ltrs

Ex:-28:- [B.Q:-8]:- $A = \frac{1}{6}$ hr's, $B = \frac{1}{8}$ hr's & C can empty 5 ltrs per min. tank filled in 12 hr's if all the pipes opened together. Find capacity of the tank?

$\frac{1}{6} + \frac{1}{8} - \frac{1}{x} = \frac{1}{12} \Rightarrow \frac{4x+3x-24}{24x} = \frac{1}{12}$

$4x + 3x - 24 = 2x$

$\Rightarrow 7x - 24 = 2x \Rightarrow 7x - 2x = 24$

$\Rightarrow x = \frac{24}{5} = 4\frac{4}{5}$ hr's $= 4$ hr's 48 mins

C can empty the tank in $4\frac{4}{5}$ hr's with 5 ltrs per minutes. [4 hr's in min $= 4 \times 60 = 240$ minutes.]

Capacity of tank $=$

$= [240 + 48]$ min $= 288 \times 5$ litres

$= 1440$ ltrs